

Exam 4

- ① Common alloying elements for F/M steels are Cr (at about 9-12%) and Mo. Cr content supports the formation of martensitic phases, increases hardness, and prevents oxidation, but excessive Cr will reduce toughness. Mo content, along with similar elements, provides solid solution ^(SSS) strengthening. Another consideration is Si content: too much will cause the agglomeration of carbides, but too little weakens corrosion performance. Overall, when determining composition you need to consider the stability of the ferrite/martensitic phases as well as alloys that will induce SSS and precipitation hardening.

- ② Ferritic steels swell significantly less than austenitic steels for several reasons:
1. Interstitials have a larger strain field, thus they more strongly attract vacancies, and there is more recombination.
 2. Vacancies are more mobile due to lower migration barriers.
 3. Carbide precipitates impede vacancy mobility.
 4. Solutes act as strong sinks, attracting and trapping vacancies.
 5. The complex microstructure of ferrite has a high density of natural sinks.

- ③ DBTT is the temperature at which a normally ductile material becomes brittle at sufficiently low temperatures. As dose increases, irradiation embrittlement (IE) occurs, and the DBTT increases. IE is largely affected by alloy composition. For example, C is needed to stabilize the martensitic phase, but it also results in a high concentration of precipitates, which induce IE.

- ④ Fe/Cr/Al alloys, like other ferritic alloys, has favorable void swelling behavior. They also have good high temperature corrosion behavior due to the formation of an aluminum oxide layer.

Too much Cr/Al will result in unfavorable phases / intermetallics that cause embrittlement, but sufficient concentration is necessary for corrosion resistance and to stabilize ferritic bcc phase.

- ⑤ Oxide particles prevent ^{excess} swelling by trapping vacancies / gas atoms, resulting in smaller, dispersed voids / bubbles. Oxide particles also act as sinks for point defects.

⑥ Advantages of Ni alloys include corrosion resistance and good high temperature properties, making them good candidates for low dose components in LWRs. However, they can be susceptible to SCC and helium generation through the neutron capture of Ni^{58} , leading to helium embrittlement.

⑦ USi fuel has a higher uranium loading, allowing for use of lower enriched uranium. However, USi fuel exhibits much higher gaseous FP swelling behavior. U_3Si exhibits more ^{and unstable} swelling, and the bubbles are heterogeneous (varying sizes). U_3Si_2 exhibits less and more stable swelling, and it is more homogeneous in nature.

⑧ The gamma phase of U-Mo is not the thermodynamically stable phase, but there are also kinetic factors at play. Irradiation dose allows the gamma phase to remain stable, and there exists a minimum fission rate to balance the thermodynamic driving force.

⑨ The solidus / liquidus gap means that during cooling, Mo solidifies first. This can result in "pockets" of Mo during fabrication that lead to inhomogeneous concentrations in the fuel.

⑩ Fission gas swelling occurs in two phases in U-Mo fuel. At low burnups, there are a small number of bubbles along grain boundaries (GBs), but the fission gas primarily exists in a superlattice of very small bubbles that do not cause significant swelling. As burnup increases, grain refinement begins to occur, nucleating smaller grains. The fission gas in the superlattice is swept to new GBs, forming larger bubbles, until eventually, large bubbles exist throughout the fuel.

⑪ Unlike U-Mo, the U-Mo-Al interaction layer (IL) will go amorphous, facilitating pore production. Excessive Al in the IL supports this phenomenon, so a mitigation strategy is to add Si to help prevent the $UAl_3 + Al \rightarrow UAl_4$ reaction. Other mitigation strategies are to coat the fuel particles, change the matrix material, or abandon dispersion fuel altogether with monolithic fuel.

- ⑫ There are numerous benefits to Al cladding. Al is cheap, widely available, well known, easy to manufacture, highly ductile, and exhibits very high thermal conductivity. However, it is only really suitable for the low temperature environment of research reactors due to its very low melting point. Even in research reactors, the operating temperature is about $\frac{1}{3}$ of the melting point.